

Checkpoint 6: Capacity Control and Generalization

Objective

Check parameter-count effects and evaluate qualitative behavior on unpaired external inputs.

Key Message

The capacity-control result makes the final claim honest: residual is not best on every metric, but remains best on SSIM, Delta E, and entropy.

Completed Work

- Completed parameter-matched U-Net capacity control with base_channels=42.
- Compared parameter-matched U-Net with the residual model.
- Ran qualitative inference on unpaired turbid images and video-derived frames.

Evidence

Metric	Matched U-Net	Residual	Better
MSE	0.034294	0.035727	Matched U-Net
PSNR	16.356273	16.304393	Matched U-Net
SSIM	0.746291	0.788963	Residual
Delta E	27.225774	26.840257	Residual
Entropy	4.790907	4.368050	Residual

2.jpg turbid



2.jpg restored



2.jpg clear



8.jpg turbid



8.jpg restored



8.jpg clear



13.jpg turbid



13.jpg restored



13.jpg clear



26.jpg turbid



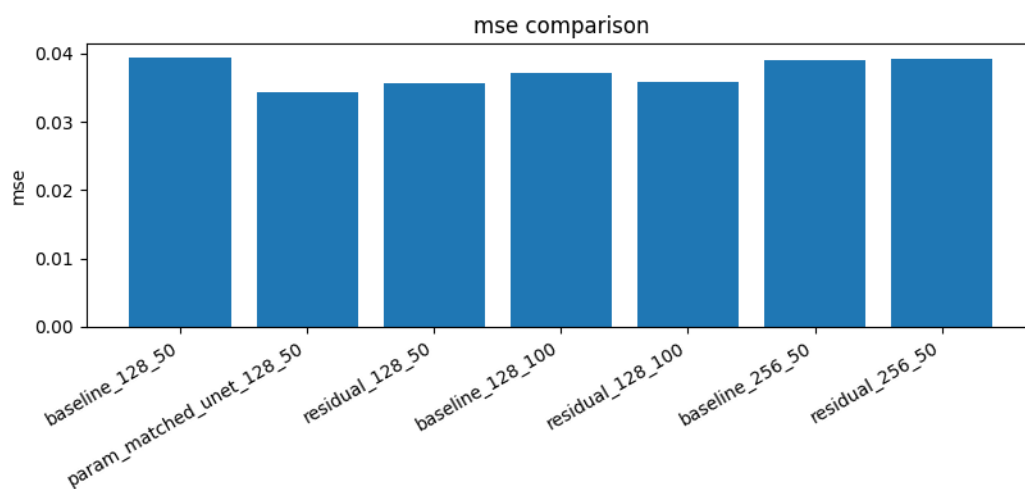
26.jpg restored



26.jpg clear



Parameter-matched U-Net examples.



MSE comparison showing the capacity-control effect.

Notes

- External datasets are qualitative only because paired clear references are not available.